

2.1 Biomolecules

Biomolecules are the most essential organic molecules, which are involved in the maintenance and metabolic processes of living organisms. These non-living molecules are the actual foot-soldiers of the battle of sustenance of life. They range from small molecules such as primary and secondary metabolites and hormones to large macromolecules like proteins, nucleic acids, carbohydrates, lipids etc.

2.2 Types of Biomolecules

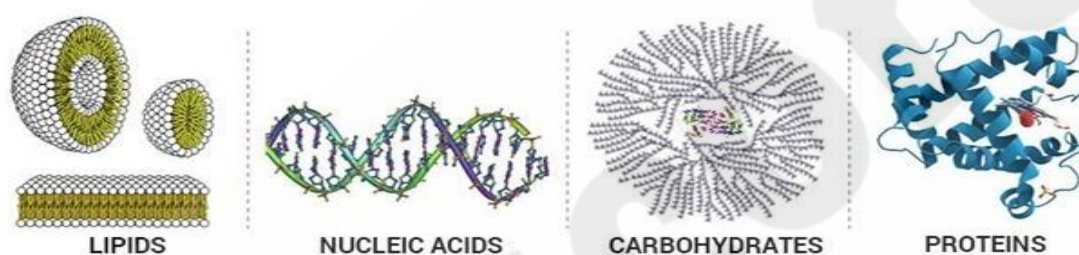


Fig. 2.1 Types of Biomolecules

There are four major classes of Biomolecules – Carbohydrates, Proteins, Nucleic acids and Lipids. Shown in Fig. 2.1. Each of them is discussed below.

2.3 Carbohydrates

Carbohydrates are chemically defined as polyhydroxy aldehydes or ketones or compounds which produce them on hydrolysis. In layman's terms, we acknowledge carbohydrates as sugars or substances that taste sweet. They are collectively called as saccharides (Greek: sakcharon = sugar). Depending on the number of constituting sugar units obtained upon hydrolysis, they are classified as monosaccharides (1 unit), oligosaccharides (2-10 units) and polysaccharides (more than 10 units). They have multiple functions' viz. they're the most abundant dietary source of energy; they are structurally very important for many living organisms as they form a major structural component, e.g. cellulose is an important structural fibre for plants. Carbohydrates are macronutrients and are one of the three main ways by which our body obtains its energy.

They are called carbohydrates as they comprise **carbon, hydrogen** and **oxygen** at their chemical level. Carbohydrates are essential nutrients which include sugars, fibers and starches. They are found in grains, vegetables, fruits and in milk and other dairy products. They are the basic food groups which play an important role in a healthy life.

The food containing carbohydrates are converted into glucose or blood sugar during the process of digestion by the digestive system.

Our body utilizes this sugar as a source of energy for the cells, organs and tissues. The extra amount of energy or sugar is stored in our muscles and liver for further requirement. The term ‘carbohydrate’ is derived from a French term ‘**hydrate de carbone**’ meaning ‘hydrate of carbon’. The general formula of this class of organic compounds is $C_n(H_2O)_n$. carbohydrates are shown in Fig. 2.2.



Fig. 2.2 carbohydrates food

2.3.1 Classification of Carbohydrates

The carbohydrates are further classified into simple and complex which is mainly based on their chemical structure and degree of polymerization.

A. Simple Carbohydrates (Monosaccharides, Disaccharides and Oligosaccharides)

Simple carbohydrates have one or two sugar molecules. In simple carbohydrates, molecules are digested and converted quickly resulting in a rise in the blood sugar levels. They are abundantly found in milk products, beer, fruits, refined sugars, candies, etc. These carbohydrates are called empty calories, as they do not possess fiber, vitamins and minerals.

Plants, being producers, synthesize glucose ($C_6H_{12}O_6$) using raw materials like carbon dioxide and water in the presence of sunlight. This process of photosynthesis converts solar energy to chemical

energy. Consumers feed on plants and harvest energy stored in the bonds of the compounds synthesized by plants.

1. Monosaccharides

Glucose is an example of a carbohydrate monomer or monosaccharide. Other examples of monosaccharides include mannose, galactose, fructose, etc. The structural organization of monosaccharides is as follows: (Fig. 2.3)

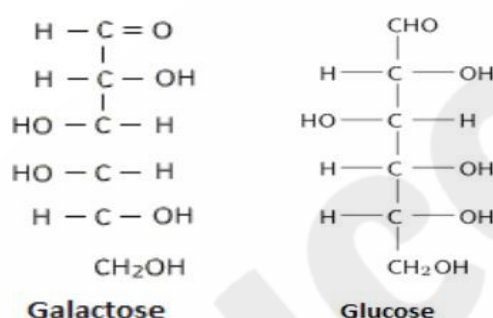


Fig. 2.3 structural organization of monosaccharides

Monosaccharides may be further classified depending on the number of carbon atoms:

- (i) **Trioses ($C_3H_6O_3$)**: These have three carbon atoms per molecule. Example: Glyceraldehyde
- (ii) **Tetroses ($C_4H_6O_4$)**: These monosaccharides have four carbon atoms per molecule. Example: Erythrose.
- (iii) **Pentoses,**
- (iv) **Hexoses, and**
- (v) **Heptoses**

2. Disaccharides

Two monosaccharides combine to form a disaccharide. Examples of carbohydrates having two monomers include- Sucrose, Lactose, Maltose, etc. structural organization is shown in Fig. 2.4.

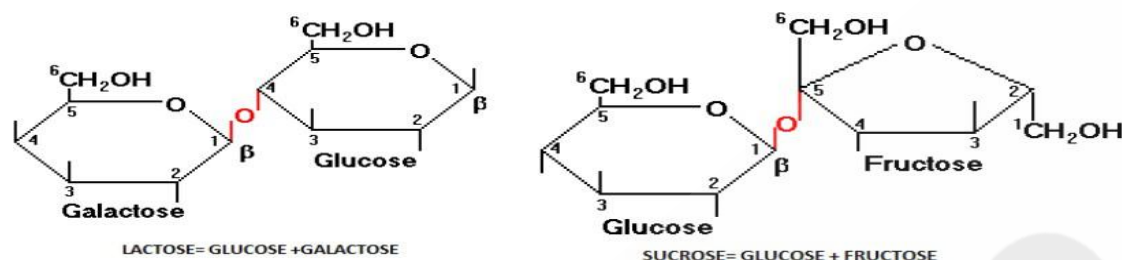


Fig. 2.4 structural organization of two monosaccharides

3. Oligosaccharides

Carbohydrates formed by the condensation of 2-9 monomers are called oligosaccharides. By this convention, trioses, pentoses, hexoses are all oligosaccharides.

B. Complex Carbohydrates (Polysaccharides)

Complex carbohydrates have two or more sugar molecules, hence they are referred to as starchy foods. In complex carbohydrates, molecules are digested and converted slowly compared to simple carbohydrates. They are abundantly found in lentils, beans, peanuts, potatoes, peas, corn, whole-grain bread, cereals, etc.

Polysaccharides are complex carbohydrates formed by the polymerization of a large number of monomers. Examples of polysaccharides include starch, glycogen, cellulose, etc. which exhibit extensive branching and are homopolymers – made up of only glucose units. Shown in Fig. 2.5.

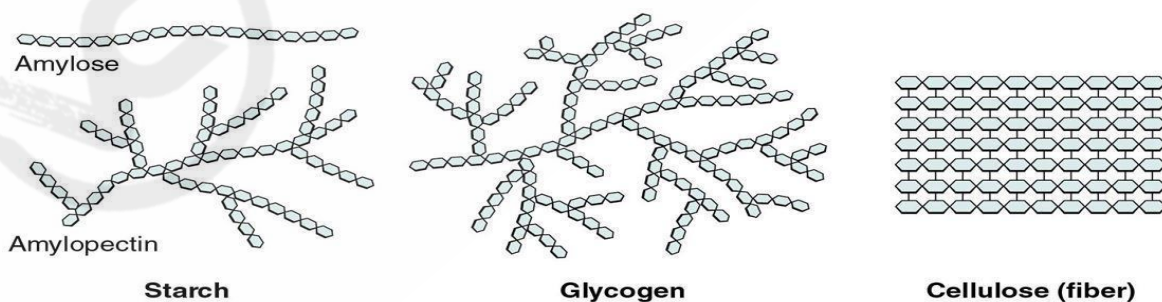


Fig. 2.5 Polysaccharides

1. Starch is composed of two components- amylose and amylopectin. Amylose forms the linear chain and amylopectin is a much-branched chain.

2. Glycogen is called animal starch. It has a structure similar to starch, but has more extensive branching.
3. Cellulose is a structural carbohydrate and is the main structural component of the plant cell wall. It is a fibrous polysaccharide with high tensile strength. In contrast to starch and glycogen, cellulose forms a linear polymer.

2.3.2 Functions of Carbohydrates

The main function of carbohydrates is to provide energy and food to the body and to the nervous system.

1. Carbohydrates are known as one of the basic components of food, including sugars, starch, and fibre which are abundantly found in grains, fruits and milk products.
2. Carbohydrates are also known as starch, simple sugars, complex carbohydrates and so on.
3. It is also involved in fat metabolism and prevents ketosis.
4. Inhibits the breakdown of proteins for energy as they are the primary source of energy.
5. An enzyme by name amylase assists in the breakdown of starch into glucose, finally to produce energy for metabolism.

1.3.3 Sources of Carbohydrates

1. Simple sugars are found in the form of fructose in many fruits.
2. Galactose is present in all dairy products.
3. Lactose is abundantly found in milk and other dairy products.
4. Maltose is present in cereal, beer, potatoes, processed cheese, pasta, etc.
5. Sucrose is naturally obtained from sugar and honey containing small amounts of vitamins and minerals.

These simple sugars that consist of minerals and vitamins exist commonly in milk, fruits, and vegetables. Many refined and other processed foods like white flour, white rice, and sugar, lack important nutrients and hence, they are labelled “**enriched**.” It is quite healthy to use vitamins, carbohydrates and all other organic nutrients in their normal forms.

2.3.4 Carbohydrate Foods

Eating too much sugar results in an abnormal increase in calories, which finally leads to obesity and in turn low calories leads to malnutrition. Therefore, a well-balanced diet needs to be maintained to have a healthy life. That is the reason a balanced diet is stressed so much by dietitians.

Table 1.1 Differences between the good and bad carbohydrates.

Good Carbohydrates	Bad Carbohydrates
High in Nutrients	Low in nutrients
Moderate in calories	High in calories
Low in sodium and saturated fats	High in sodium and saturated fats
Low in trans-fat and cholesterol	High in trans-fat and cholesterol
They are complex carbs. For instance: Legumes, vegetables, whole grains, fruits, and beans.	Foods considered bad carbs rarely have any nutritional value. Some of the foods include white flour, rice, pastries, sodas and processed foods.

Examples of Carbohydrates

Following are the important examples of carbohydrates:

- Glucose
- Galactose
- Maltose
- Fructose
- Sucrose
- Lactose
- Starch
- Cellulose
- Chitin

2.4 PLA Bioplastics

Polylactic acid plastic, also named PLA plastic, is a type of polymer derived from renewable sources like tapioca roots, cornstarch and sugarcane. Its properties are similar to synthetic plastic, and it is eco-friendly. Hence it entered the plastic industry as a competitive commodity.

2.4.1 How are PLA bioplastics made?

Polylactic plastic is made by fermenting carbohydrate sources like sugarcane, maize or corn starch under controlled conditions. The building blocks of PLA plastic, which is lactide monomer or lactic acid, are later polymerized into PLA.

In the beginning, the starch gets separated when sugarcane goes through wet milling. Afterwards, it gets mixed with enzymes or acid and then it is heated. The heating procedure converts starch into corn sugar or Dextrose. Finally, the derived sugar produces L- Lactic acid, the fundamental component of PLA plastic.

2.4.2 Types of PLA bioplastics

There are many types of PLA plastics, but few important ones are regular Poly L-lactide (PLLA), racemic Poly L-lactic Acid (PLLA), Poly DL- Lactic Acid (PDLLA) and Poly D-lactic Acid (PLDA).

Though all the types of PLA plastics are made up of the same renewable source – lactic acid, the orientation of atoms are different in these types. Also, the physical and chemical properties slightly vary in all of them.

2.4.3 Uses of PLA bioplastics

PLA bioplastic has a wide range of applications in various sectors like textiles, packaging, medical, cosmetics and so on. Some of the uses of PLA plastic are:

i) Medical

Since PLA bioplastics are biodegradable and biocompatible, it has vital importance in the medical industry. The healthcare industry uses polylactic acid plastic in manufacturing tissue engineering stages and bone fixation devices like plates, screws, rods, pins, surgical equipment and meshes. Its adaptability with minimum infection and inflammation has membrane applications like covering wounds and making stitches plastic. Besides, PLA plastics are also used to deliver medical

materials, medical implants and are used in various dermatological procedures like scar rejuvenation and facial lipoatrophy.

ii) Textiles

In the textiles and apparel industry, efforts have been made to replace synthetic polyester textiles with renewable textiles. Due to its lightweight, recyclability, good resistance to sun rays, and breathability, PLA plastic is used to make carpets, beddings, shirts and sportswear. Since PLA plastic has low smoke generation characteristics, it is also used to manufacture insulating materials, garbage bags, and protective foams.

iii) Cosmetics

As a result of consumer awareness regarding synthetic plastic pollution, cosmetic industries are now shifting towards PLA biodegradable plastics. Products like cotton buds, razors, hairbrushes, toothbrushes and diapers are made using polylactic acid plastics.

iv) Packaging and disposable

PLA bioplastics are typically safe for packaging food, and it is also approved by the Food and Drug Administration (FDA). Though it can discharge lactic acid when it comes in contact with certain liquids, its concentration does not harm humans.

In the form of rigid thermoforms, PLA bioplastic is used to manufacture yoghurt pots, meat trays, bakery, herbs and candy containers, coffee lids, cold drink cups, fruit and vegetable clamshells. In the bottling sector, it is used to make milk bottles with a short shelf life, edible oil bottles and mineral water bottles.

As films, it is used as a wrap for flowers and candy twists, making lamination films, display carton windows, tapes, bread bags, cake mix pouches, die-cut labels, and envelopes.

v) Structural uses

Due to its durability, high heat resistance and hydrolytic stability, biobased PLA plastic is used in making structures for electrical and electronic appliances.

It has significant use in the automobile industry and is used for manufacturing high-end cars' wheel covers, translucent roofs, under the hood parts and safety helmets.

PLA bioplastics are also suitable for making pillow covers, floor mats, door trims, ceiling material and front panels due to their superior tensile strength.

2.4.4 Advantages of PLA Bioplastics

PLA plastic has many benefits as compared to competing polymers, and that's why manufacturers prefer it. Some of them are:

- PLA plastics are derived from biomass, thus it is biodegradable and more environmentally friendly.
- As compared to other plastic production, PLA plastics require less energy to produce.
- When polylactic plastic degrades, it transforms into natural materials like carbon-di-oxide, water which reduces plastic waste.
- Since PLA is a thermoplastic, it can be melted, recycled and reshaped without changing its chemical properties.
- It is carbon neutral, which means it doesn't emit carbon dioxide into the atmosphere when recycled.
- PLA is non-toxic, that's why it is used in the food packaging and medical industry.
- When oxygenated, it does not produce any toxic fumes as synthetic plastics do.
- It has a smoother, shinier and glossy appearance; hence it is used in textiles and automobile industries.
- PLA plastics emit a sweet smell in the printing process and can easily be printed even on a cold surface.
- As compared to other bioplastics like polyhydroxyalkanoate (PHA), poly γ -caprolactone (PCL) and polyethylene glycol (PEG), it has better thermal manufacturability.

2.4.5 Disadvantages of PLA Bioplastics

Though PLA plastics have many applications and unique properties it also has some disadvantages like:

- If PLA plastics do not get the ideal temperature and pressure, it does not degrade and takes around 100 to 1000 years to decay.

- As compared with other bioplastics, PLA has a lower glass transition temperature of 55 degree Celsius.
- Unless mixing PLA with other polymers, it is primarily more brittle and ductile.
- PLA bioplastics also have a lower crystallization rate and strength as compared to petroleum-based plastics.
- Mixing PLA bioplastics with synthetic plastic could contaminate the recycling procedure as it has different chemical properties. Hence, during the recycling process, it should be separated from traditional plastic.
- It is thermally unstable and also has poor gas barrier characteristics.
- It has a high production cost.

2.4.6 Properties of PLA Plastics

Due to its similar properties as petroleum-based plastics, PLA plastic is a great biodegradable alternative. Some of the characteristics of Polylactic acid plastics are:

i) Thermal Properties

The thermal properties of PLA bioplastic usually depend upon its molecular weight, thus, it is classified as a semi-crystalline polymer. It has a high melting point of 180 degree Celsius but a low glass transition temperature of 55 degree Celsius. It is ultraviolet rays resistant and has low smoke formation and flammability during its production.

ii) Mechanical properties

PLA plastics have high mechanical strength and resistance with a good appearance compared to other polymers like polyurethane, polystyrene and polypropylene. The tensile strength, flexural strength and 'Elongation at break' of PLA plastics are also better than other bioplastics.

iii) Rheological properties

PLA bioplastic is non-Newtonian fluid and pseudoplastic, which means the change of viscosity depends upon the stress applied to it. Also, when tension is applied to it, its viscosity decreases.

2.4.7 Is PLA plastic sustainable?

In a broader sense, PLA plastics are sustainable as it comes from renewable sources. However, PLA plastics require precise temperature and pressure to degrade. In the degradation process of PLA

plastic first, it needs to be heated at 60 degree Celsius. Then it should be exposed to particular microorganisms which digest the plastic and decompose it.

Secondly, PLA plastics need to be sorted out when recycling as they can't be recycled with other plastics due to their different chemical properties. This increases the cost of transportation for bioplastic companies.

2.5 Polyhydroxyalkanoate (PHA)

Biodegradable plastic called PHA, which stands for Polyhydroxyalkanoate, is a type of polymer produced by bacteria. Therefore, it can easily disintegrate in the environment. Certain bacteria produce PHA when they are under stressed conditions. PHA can be used in many different applications. They can also be modified and improved to suit our needs. Each year more and more types of bacteria producing PHA are identified.

In present times we use plastic almost everywhere and therefore it is important to replace plastic which is non-degradable with biodegradable, such as PHA. Plastic is not only used to produce plastic bags, disposable forks, spoons or PET plastic bottles. Very important field is also the medical and pharmaceutical field. There is also a possibility to use PHA in agriculture. Decomposition in nature of non-biodegradable plastics can take between hundreds of years to thousands of years. It is also one of the largest sources of pollution in oceans. Plastics that are not disposed properly pollute not only water but also air and soil. Some plastics are burned and this produces toxins in the air which can cause respiratory problems but also neurological diseases. Also animals suffer when they eat pieces of plastic and die due to this. Plastic can degrade by ultraviolet light from the sun into very small particles called microplastics. They are also dangerous because they are too small to be collected but they do harm to animals. It is predicted that by 2050 there will be more plastic in the oceans than fish.

2.5.1 Microplastics in water

i) What are microplastics?

Small pieces of plastic from disintegrated bigger plastic objects.

ii) How does microplastic affect the ocean?

Aquatic life and birds can mistake microplastics for food. Also microplastics can absorb harmful chemicals and they can contaminate water or make harm to aquatic animals.

iii) Toxic air

Toxic air can cause problems with babies' brain development with lifelong implications, children's learning and development. Pollutants are harmful for babies' lungs development. For adults toxic air can cause respiratory problems and even be a cause of death.

2.5.2 Is PHA a biodegradable plastic?

Yes PHA is biodegradable because it is created naturally inside bacteria and also PHA plastic can be degraded naturally by microorganisms. PHAs are synthesized by many different bacteria as carbon and energy storage capsules.

PHA was discovered for the first time in 1926 in one type of bacterium called *Bacillus megaterium*. First efforts to use PHB (a type of PHA) commercially was in the 1950s. However, the process of PHB production was not so easy and unprofitable. Later in 1970 a company called Biopol started production of PHA. Later other companies also started the production process. Since this time there have been over 150 different types of PHAs discovered. PHA plastics can also be modified in order to make them more stiff or brittle.

PHAs are not only biodegradable plastics but also thermoplastics. Their melting temperature is between 50°C and 180°C. PHAs molecules are not toxic and can be used in medicine but it is necessary to check if they are biocompatible. It means that when they are introduced to a human's body they should not cause severe immune reactions.

2.5.3 What can PHA plastic be used for

i) Medical applications:

- **Cardiovascular tissue engineering** – PHAs can be used instead of synthetic materials. Bioplastics can be used to repair or replace damaged, diseased, or malfunctioned blood vessels. Scaffolds in tissue engineering can also be made out of PHAs. They get covered with cells and can be implanted for a few months. No medical problems or inflammatory responses were observed. Heart valves can also be produced using PHAa.

- **Bone marrow scaffolds.** They can be used for patients with bone loss. Biodegradable scaffolds can be covered with bone marrow cells. With time the biodegradable scaffold will disintegrate and bone marrow cells will stay in the suitable place.
- **Cartilage repair** – bioplastic can be used in cartilage tissue engineering to regenerate cartilage.
- Biodegradable plastic can also be used in **eye disorders** to create scaffolds used in eyelid reconstruction.
- PHAs can be used as scaffolds for **nerve regeneration**.
- **Wound management, skin regeneration** – wound dressing or scaffolds. PHAs are also a good material to be used in reconstructive medicine to treat injured skin. Special scaffolds can be created that are comparable to those of human skin. Special cells can be seeded on these scaffolds. They can be used as a matrix material for skin tissue engineering. Such scaffolds can be used in difficult contours of the body. Also wounds covered with specially prepared film can heal more quickly. The membrane can also reduce inflammation.
- From the 1990s PHAs were studied in **drug delivery systems**. Therapeutics can be delivered for prolonged release. As a natural polymer they are biodegradable and biocompatible.
- New **tissue formation or regeneration**.
- Development of **artificial esophagus**.
- Biodegradable plastics can be used as absorbable fibers, surgical meshes, repair patches, suture material, suture fasteners, screws, bone plates, bone filling augmentation material and many more.
- PHAs are also great to be used in **pediatric applications** because when they are applied for children they do not hinder growth.
- They can also be used for **blood vessels repair, heart valve scaffold**.
- In cancer therapy and **controlled drug delivery**.
- **Orthopedic applications** of PHAa include bone accumulation or maintenance of osteoblast phenotype.
- **Cartilage engineering**

ii) Non-medical applications

- **Packagings** that can be degraded into carbon dioxide and water by microorganisms. From PHA bottles, bags and films can be produced. Already in the 1980s Biopol started experimenting with packagings made of PHA. First packagings were shampoo bottles, parts of sanitary products for women and diapers that created moisture barriers, shopping bags, containers, cups, medical surgical garments, upholstery, carpets.
- **Fiber** material which possibly could be used in textile industry
- **Biofuels** which can be produced low cost
- PHB can be used as **paper finishing** to coat paper as they are water insoluble, can create oxygen barrier and are UV resistant. They also have excellent film-forming properties. Paper covered with PHB becomes more water resistant.
- Agriculture applications of PHA as a carrier for targeted and controlled **delivery of pesticides to soil**. PHA becomes later degraded in the soil by microbes. Thanks to this method herbicides do not impact so much non-targeted species and also reduce the need for repeated applications. PHAs can help with controlling bacterial pathogens. Addition of PHB in the feed of prawns reduced the amount of bacteria and therefore the survival rate of the prawn larvae is higher and their development is better.
- Silver **nanoparticles** which are known for their antibacterial properties can be **stabilised** by addition of PHAs. Without it they are not stable and therefore their use is limited.

2.6 cellulose-based water filters

Filtration is a common way to obtain pure drinking water by removing particles and microorganisms based on size exclusion. Cellulose-based filters are affordable and biobased option for the removal of particles but bacteria are usually too small to be removed by size exclusion alone. In this article, the surfaces of cellulose fibres in two types of commercial paper filters have been given a positive net charge to trap bacteria through electrostatic interactions without releasing any biocides. The fibres were modified with the cationic polyelectrolyte polyvinylamine polymer in single layers (1 L) or in multilayers together with the anionic polyelectrolyte polyacrylic acid (3 L or 5 L) using a water-based process at room temperature. Filtration tests show that all filters, using both types of filter papers and a number of layers, can physically remove more than 99.9% of *E. coli* from water and that the 3 L modified filters can remove more than 97% of cultivatable bacteria from natural

water samples. The bacterial reduction increased with increasing number of filter sheets used for the filtration and the majority of the bacteria were trapped in the top sheets of the filter. The results show the potential for creating water purification filters from bio-based everyday consumable products with a simple modification process. The filters could be used in the future for point-of-use water purification that may be able to save lives without releasing bactericides. (Fig. 2.6)



Fig. 2.6 Filtration

2.6.1 Methods

A. Nitrogen analysis

A nitrogen analysis was performed using an ANTEK MultiTek analyzer (PAC, Houston, TX, USA) to quantify the amount of PVAm adsorbed onto the paper filters. Approximately 5 mg of the material was analysed per measurement and an average of three measurements per sample was calculated.

B. Flow rate for free flow filtration

The flow rate for free flow filtration through the paper filters due to gravity was measured by filtering dH₂O through different numbers of sheets of the reference and 3 L LbL modified filters, assembled in syringe filter holders. The time taken to filter 10 mL of water was measured and the flow rate was calculated as mL min⁻¹. The reported value is the average flow rate of three measurements.

2.7 Proteins

Proteins are another class of indispensable biomolecules, which make up around 50 per cent of the cellular dry weight. Proteins are polymers of **amino acids** arranged in the form of polypeptide chains. The structure of proteins is classified as primary, secondary, tertiary and quaternary in some cases. These structures are based on the level of complexity of the folding of a polypeptide chain. Proteins play both structural and dynamic roles. Myosin is the protein that allows movement by contraction of muscles. Most enzymes are proteinaceous in nature.

2.7.1 Protein Structure

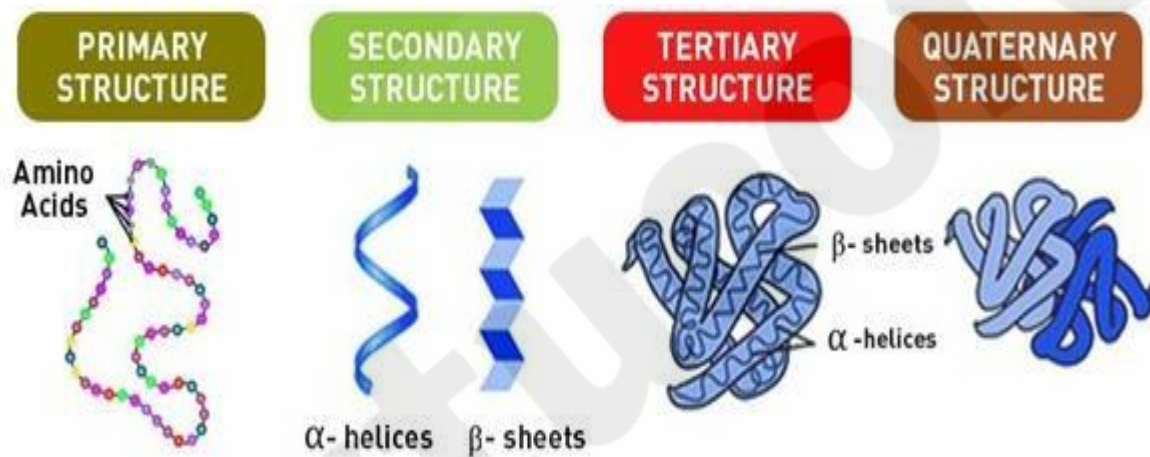


Fig. 27 Protein Structure

In general, there are two types of protein molecules: fibrous proteins and globular proteins. Fibrous proteins are insoluble and elongated. Globular proteins are soluble and compact. Fibrous and globular proteins may comprise one or four types of protein structures and they include primary, secondary, tertiary, and quaternary structure. (Fig. 2.7)

Primary Structure: It is a specific sequence of amino acids. The order of amino acids bonded together is determined by information stored in genes.

Secondary Structure: It is a three-dimensional form of a local segment of proteins. They are formed by hydrogen bonds between the atoms along the backbone of the polypeptide chain.

Tertiary Structure: It is determined by R-groups. It is a three-dimensional shape of a protein. Many numbers of tertiary structure fold to form Quaternary Structure.

Quaternary Structure: It is the arrangement of multiple folded protein subunits in a multi-subunit complex.

2.7.2 Types of Proteins and Their Functions

Although there are debates about the intake of carbohydrates and fats in order to maintain a proper health, a minimum amount of daily protein intake is always a doctor's first recommendation. The common examples of proteins in biology are eggs, almond, chicken, oats, fish and seafood, soy, beans and pulses, cottage cheese, Greek yogurt, milk, broccoli, and quinoa.

Functions of Proteins

1. **Enzymes:** Enzymes mostly carry out all numerous chemical reactions which take place within a cell. They also help in regenerating and creating DNA molecules and carry out complex processes.
2. **Hormones:** Proteins are involved in the creation of various types of hormones which help in balancing the components of the body. For example hormones like insulin, which helps in regulating blood sugar and secretin. It is also involved in the digestion process and formation of digestive juices.
3. **Antibody:** Antibody also known as an immunoglobulin. It is a type of protein which is majorly used by the immune system to repair and heal the body from foreign bacteria. They often work together with other immune cells to identify and separate the antigens from increasing until the white blood cells destroy them completely.
4. **Energy:** Proteins are the major source of energy that helps in the movements of our body. It is important to have the right amount of protein in order to convert it into energy. Protein, when consumed in excess amounts, gets used to create fat and becomes part of the fat cells.

Table 2.2 Functions of Proteins.

Aspect	Functions of Proteins in Human Body	Examples
Storage	Legume Storage, albumin, and proteins.	Supplies food during the early stage of the seedling or embryo.

Hormone Signalling	Counterpart activities of different body parts.	Glucagon and Insulin.
Transport	It transport substances throughout the body through lump or blood cells.	Hemoglobin.
Contraction	To carry out muscle contraction.	Myosin.
Digestive Enzyme	Breaks down nutrients present in the food into smaller portions so that it can be easily absorbed	Pepsin, Amylase, and Lipase

2.7.2 Introduction to Whey Proteins

Whey proteins are easily and quickly digested. Whey protein is one of the 2 proteins of cow's milk. Whey proteins refer to the individual protein which is separated out from the casein while cheesemaking. These proteins are purified into different concentrations based on the end composition desired. The content of the whey protein may vary in carbohydrates, fat, immunoglobulins, lactose and minerals. Whey proteins are loaded with EAAs (Essential Amino Acids) including three BCAAs (Branched Chain Amino Acids) and also they contain subcomponents of micro fractions which provide the benefits of elemental nitrogen and amino acids. Whey protein is the one which is most useful for sports nutrition. In addition to these, whey products are evident in salad dressings, infant formulas, emulsifiers, baked goods and medical nutritional formula.

VARIETIES OF WHEY PROTEIN There are three main types of whey protein which is obtained from various processing techniques that is used to separate whey protein.

They are

- Whey powder,
- Whey concentrate and
- Whey isolate.

2.7.3 Benefits of Whey Protein

Whey protein is considered as an excellent protein for the choice of individuals of all ages for healthy diet and also to improve and maintain their health. Traditionally, Whey protein was only

used by most of the athletes and bodybuilders to promote the muscle growth . But from past few years, whey protein is being used in some other applications. Some applications that using whey proteins are: cancer treatment, wound healing, infant health and weight loss.

Benefits of Whey protein may include:

- Whey protein helps to increase the serotonin activity and helps to promote restful sleep ;
- Whey protein helps enhance energy levels; it helps to decrease the stress.
- It helps to keep the metabolic rate high.
- It helps to reduce body fat and build the lean body mass.
- It helps to improve the memory loss under stress.

In addition to these, some of the top benefits of whey protein may include:

it provides immunity support, increase muscle mass, boost metabolism, and helps to improve overall health

Whey proteins promote:

- Muscle strength
- Improved immune system
- Muscle synthesis
- Performance / endurance
- More favorable body composition
- Recovery

2.7.4 Whey Protein Applications

Whey proteins are used in various industries for the supplement of food and beverages. Whey proteins are used for medicine purposes and to improve the overall health.

i) CANCER

Cancer patients are undergoing chemotherapy or radiation may have difficulty in meeting their nutritional requirements and this is because of lack of appetite. So, this may lead to muscle loss, weight loss and protein calorie malnutrition. Here, the use of whey protein provides excellent protein choice for the cancer patients . In addition to these, it helps them very easy to digest.

ii) DIABETES

Today, most of the people are affected by Type 2 diabetes and it is one of the growing health problems over the world. It becomes a concern for teenagers and children. Healthy nutrition practices may play a role in managing and helping the type-2 diabetes. Here, whey protein is the high biological, a high quality and value protein which is good choice for the people who has diabetics. Whey protein helps to control the blood glucose levels and also provides additional beneficial for weight management which is a concern for type-2 diabetics .

iii) CARDIOVASCULAR HEALTH:

Heart disease is considered as one of the most leading disease which causes death for both men and women. It is essential to maintain health and nutrition diet and also regular exercise to maintain the healthy cardiovascular system . In this way, whey protein is a part of that nutritious diet, which keeps a person healthy. High blood pressure (Hypertension) is one of the leading heart diseases and it may cause stroke. Whey protein helps to fight against the hypertension. Elevated cholesterol is one of the factors that is associated with the heart disease. Here, whey protein helps to reduce cholesterol.

2.7.5 Meat analog

The definition of meat analog refers to the replacement of the main ingredient with other than meat. It also called a meat substitute, meat alternatives, fake or mock meat, and imitation meat. The increased importance of meat analog in the current trend is due to the health awareness among consumers in their diet and for a better future environment. The factors that lead to this shift is due to low fat and calorie foods intake, flexitarians, animal disease, natural resources depletion, and to reduce greenhouse gas emission. Currently, available marketed meat analog products are plant-based meat in which the quality (i.e., texture and taste) are similar to the conventional meat.

The ingredients used are mainly soy proteins with novel ingredients added, such as mycoprotein and soy leghemoglobin. However, plant-based meat is sold primarily in Western countries. Asian countries also will become a potential market in the near future due to growing interest in this product. With the current advance technology, lab-grown meat with no livestock raising or known as cultured meat will be expected to boost the food market in the future.

Also, insect-based products will be promising to be the next protein resource for human food. Nevertheless, other than acceptability, cost-effective, reliable production, and consistent quality towards those products, product safety is the top priority. Therefore, the regulatory frameworks need to be developed alongside.

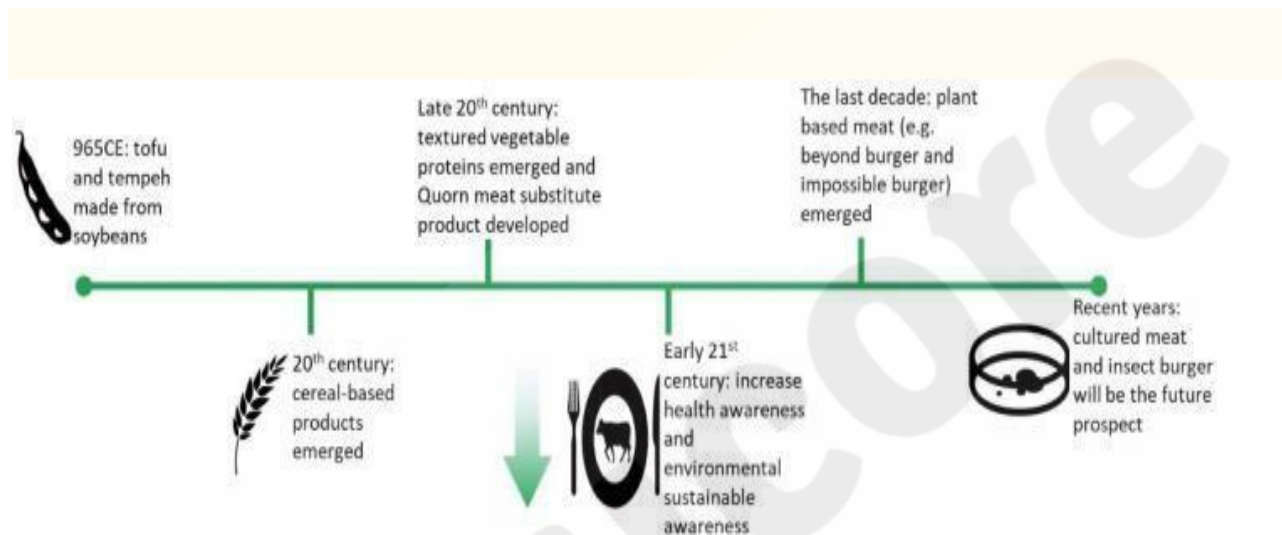


Fig. 2.8 Protein Meat analog

2.7.6 What Plants Are High in Protein?

Plant protein is simply a meaningful food source of protein which is from plants. This group can include pulses, tofu, soya, tempeh, seitan, nuts, seeds, certain grains and even peas. Pulses are a large group of plants, which include chickpeas, lentils, beans (such as black, kidney and adzuki beans) and split peas. Plant proteins are highly nutritious – not only as good sources of protein, but also because they provide other nutrients such as fibre, vitamins and minerals. Our intake of fibre tends to be too low, however by incorporating certain plant proteins into your diet, such as pulses, peas and nuts, you can easily boost your fibre intake. Did you know that our peas, soya beans and green beans are all a great source of plant fibre?

2.7.7 Why Protein Is Important?

Protein is a vital nutrient responsible for the growth, maintenance and repair of our bodies. So, we must ensure we eat enough high-quality protein every day to keep our bodies healthy. Essentially, protein is fundamental to the basic structure of our body. There are 9 essential amino acids, and when you eat a protein source containing all of these it is called a ‘complete’ protein.

Animal protein sources are complete, and they include fish, poultry such as chicken and red meat. Plant sources of protein, with the exception of soya and quinoa, are ‘incomplete’. This is because plant protein sources tend to lack at least 1 of the 9 essential amino acids. So how can you ensure that you consume all of the essential amino acids when eating plant proteins? The answer is protein combining.

2.7.8 What Is Protein Combining?

Put simply, protein combining means eating a selection of incomplete plant protein sources, to ensure you consume all of the essential amino acids. As a general rule of thumb, pulses and grains lack different essential amino acids. So, eating a pulse and a grain provides an overall complete protein source. With this in mind, it’s easy to get all the essential amino acids your body needs. For example, a serving of rice and peas or bread and hummus would give you a complete protein source. And remember, you needn’t eat your combined proteins in just one meal – they can be eaten at different points during the day or even the week!

2.7.9 How to Adopt a Plant-Based Diet?

The plant-based diet is rather simple and easy to get your head around as it’s centred around foods derived from plant sources. Plant-based diets cheer on the eating of fruits, vegetables, wholegrains, legumes, seeds and nuts, meaning these foods comprise the majority of what you eat. To give you a helping hand, our frozen vegetables are a simple yet delicious place to start if you’re looking to slowly add more plants into your lifestyle, without making the full move.

At first you may think this is a classic vegan or vegetarian diet. It’s not a plant-based diet only, it is far more flexible and doesn’t boycott dairy, meat, seafood and eggs altogether. Its easy-going and inclusive approach make this diet an attractive one – nobody likes rules, after all! And what’s more, flexitarian diets promote consuming a wide variety of plant foods, so there are of course numerous nutritional benefits to adopting this way of eating.

As a nation, we currently struggle to keep up with this figure, however with a plant-based diet heavily relying on fruit and vegetables, you’re likely to find it easier to consume the recommended amount. Likewise, as beans and pulses are greatly encouraged, you’re expected to have an increased intake of fibre, vitamins and minerals – ideal as most of us are also not consuming enough fibre! To

give you a snapshot of how humble vegetables house an array of nutritional benefits, peas provide protein, fibre, folic acid and vitamin C. It's natural to think that a plant-based diet can lead to a lack of protein, but this is far from the truth – plants provide protein. If you're trying to move towards more of a plant-based diet, now is a great time to get creative and experiment with new flavours and combinations that you've never tried before! Our Meat-Free Burgers, Sausages and Meat-free Meatballs are packed with protein and provide fibre, and each have their own tasty combo of herbs and spices.

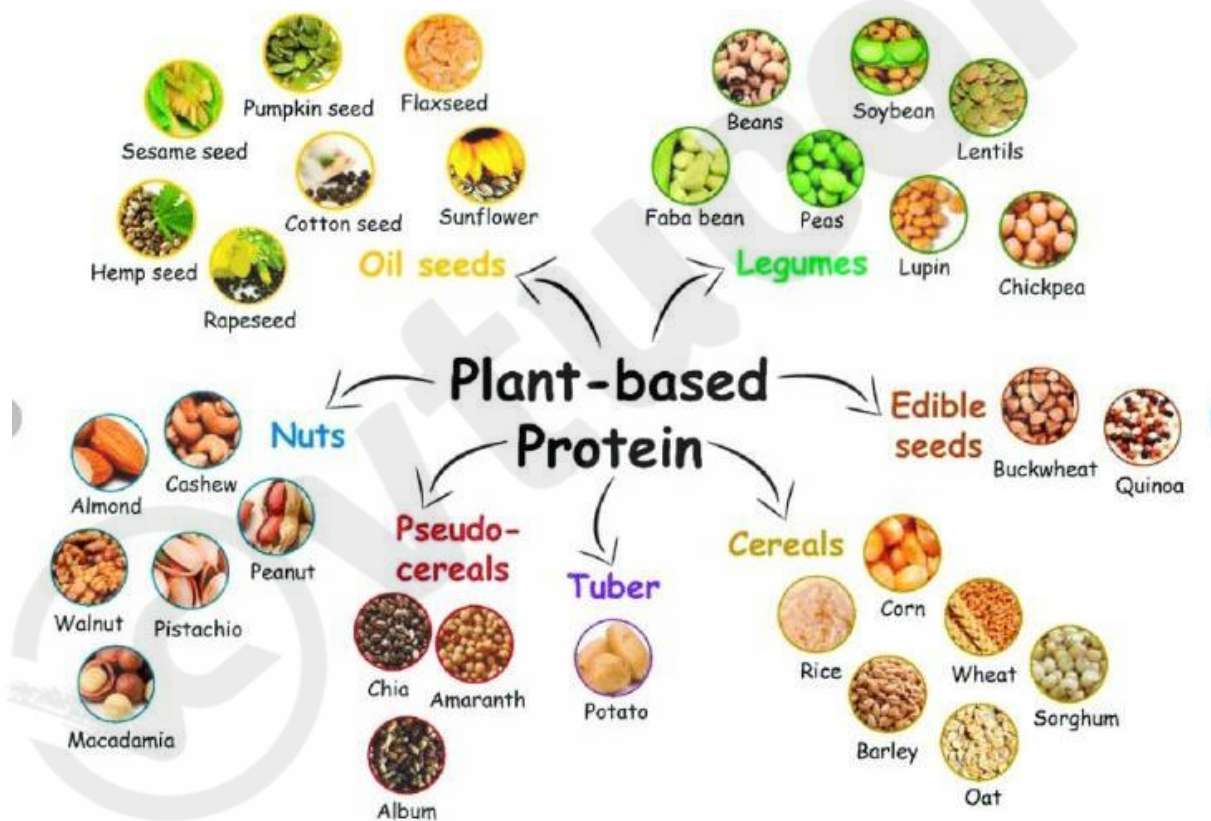


Fig. 2.9 Plant based Protein

2.8 Nucleic Acids

Nucleic acids refer to the genetic material found in the cell that carries all the hereditary information from parents to progeny. There are two types of nucleic acids namely, deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). The main function of nucleic acid is the transfer of genetic information and synthesis of proteins by processes known as translation and transcription. The monomeric unit of nucleic acids is known as nucleotide and is composed of a nitrogenous base, pentose sugar, and phosphate. The nucleotides are linked by a 3' and 5' phosphodiester bond. The nitrogen base attached to the pentose sugar makes the nucleotide distinct. There are 4 major nitrogenous bases found in DNA: adenine, guanine, cytosine, and thymine. In RNA, thymine is replaced by uracil. The DNA structure is described as a double-helix or double-helical structure which is formed by hydrogen bonding between the bases of two antiparallel polynucleotide chains. Overall, the **DNA structure** looks similar to a twisted ladder.

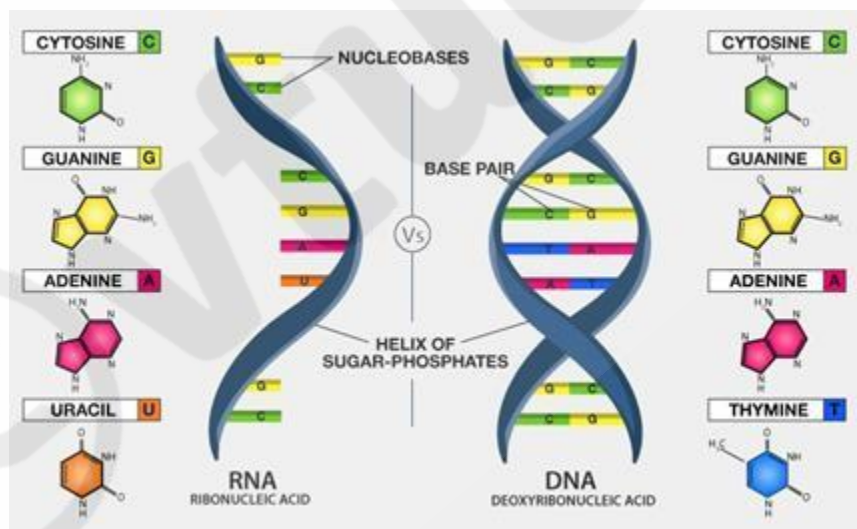


Fig. 2.10 Plant based Protein

Table 1.3 DNA and RNA Difference

DNA (Deoxyribonucleic acid)	RNA (Ribonucleic acid)
Definition	
It is a long polymer. It has a deoxyribose and phosphate backbone having four distinct bases: thymine, adenine, cytosine and guanine.	Is a polymer with a ribose and phosphate backbone with four varying bases: uracil, cytosine, adenine and guanine.
Location	
It is located in the nucleus of a cell and in the mitochondria.	It is found in the cytoplasm, nucleus and in the ribosome.
Sugar portion	
It has 2-deoxyribose.	It has Ribose.
Function	
The function of DNA is the transmission of genetic information. It acts as a medium for long- term storage.	RNA is critical for the transmission of the genetic code that is necessary for protein creation from the nucleus to the ribosome.
Predominant Structure	
DNA is a double-stranded molecule that has a long chain of nucleotides.	RNA is a single-stranded molecule which has a shorter chain of nucleotides.
Propagation	
DNA replicates on its own, it is self-replicating.	RNA does not replicate on its own. It is synthesized from DNA when required.
Nitrogenous Bases and Pairing	
The base pairing is as follows: GC (Guanine pairs with Cytosine) A-T (Adenine pairs with Thymine).	The base pairing is as follows: GC (Guanine pairs with Cytosine) A-U (Adenine pairs with Uracil).

2.8.1 DNA

- In cells, DNA (Deoxyribonucleic acid) is the nucleic acid that functions as the original blueprint for the synthesis of proteins. DNA contains the sugar deoxyribose, phosphates and a unique sequence of the nitrogenous bases adenine (A), guanine (G), cytosine (C) and thymine (T).

Brief Insight into the Structure and Composition of DNA

- The DNA molecules contain instructions a living entity requires to grow, develop and reproduce. These instructions are present inside each cell and are inherited from the parents to their offspring.
- It is made up of nucleotides which contain a nitrogenous group, a phosphate group, and a sugar group. The order of the nitrogenous bases – thymine(T), guanine(G), cytosine(C), and adenine(A), is crucial in determining the genetic code.
- Genes are formed by the order of the nitrogenous bases present in the DNA which is crucial for protein synthesis. RNA is another nucleic acid that translates genetic information into proteins from DNA.
- The nucleotides are linked together for the formation of two long strands which spiral to produce a structure known as the double-helix which resembles that of a ladder wherein the sugar and phosphate molecules form the sides while the rungs are formed by the bases.
- The bases located on one strand pair up with the bases on the other strand, as in – **guanine** pairs with **cytosine** and **adenine** pairs with **thymine**.
- The DNA molecules are extremely long and hence without the right packaging, they cannot fit into cells. Thus, DNA is tightly coiled to produce formations referred to as **chromosomes**. Every chromosome has a single DNA molecule. In humans, there are 23 pairs of chromosomes that are present within the nucleus of the cells.

2.8.2 Types of DNA

- **A-DNA:** It is found at a relative humidity of 75%. In an environment where there is a higher salt concentration or ionic concentrations, such as K^+ , Na^+ , Cs^+ or in a state of dehydration it endures in a form that contains 11 nucleotide pairs with a rise of $2.56\text{\AA}$ vertically per base pair. It has the broadest helical diameter amongst all DNA forms – $23\text{\AA}$ DNA which is a typical helix that is right-handed with a rotation of 32.7° per base pair.
- **B-DNA:** The most common form, present in most DNA at neutral pH and physiological salt concentrations, is B-form. It has 10 base pairs per turn from the helix axis. There is a distance of $3.4\text{\AA}$ with a helical diameter of $20\text{\AA}$. Watson-Crick's double helix model is defined as a B-form of DNA.
- **C-DNA:** It is observed at a relative humidity of 66% and in the occupancy of a few ions such as Lithium(Li^+). It closely has 9.33 base pairs for every turn. The diameter of the helix is about $19\text{\AA}$ and the vertical rise for every base pair for the right-handed helix is 3.320 .
- **D-DNA:** It is observed rarely as an extreme variant. The 8 base pairs are tilted negatively from the helix axis with an axial rise of about $3.03\text{\AA}$
- **Z-DNA:** It is found in an environment with a very high salt concentration. Unlike the A, B, and C types of DNA, it is a left-handed helical structure. The backbone is arranged in a zig-zag pattern formed by the sugar-phosphate linkage wherein the recurrent monomer is the dinucleotide in contrast to the mononucleotide, which is observed in alternate forms.

2.8.3 RNA

Ribonucleic acid (RNA) is a nucleic acid which is directly involved in protein synthesis. Ribonucleic acid is an important nucleotide with long chains of nucleic acid present in all living cells. Its main role is to act as a messenger conveying instructions from DNA for controlling protein synthesis.

RNA contains the sugar ribose, phosphates, and the nitrogenous bases adenine (A), guanine (G), cytosine (C), and uracil (U). DNA and RNA share the nitrogenous bases A, G, and C. Thymine is usually only present in DNA and uracil is usually only present in RNA.

2.8.4 Types of RNA

Only some of the genes in cells are expressed into RNA. The following are the types of RNA wherein each type is encoded by its own type of gene:

- **tRNA**– The transfer RNA or the tRNA carries amino acids to ribosomes while translation
- **mRNA** – The messenger RNA or the mRNA encodes amino acid sequences of a polypeptide
- **rRNA** – The ribosomal RNA or the rRNA produces ribosomes with the ribosomal proteins that are organelles responsible for the translation of the mRNA.
- **snRNA** – The small nuclear RNA forms the complexes along with proteins which are utilized in RNA processing in the eukaryotes.
- With an estimated global mortality of about 50,000 per year, rabies has been identified as one of the major causes of human death from infectious diseases. According to estimates, one person dies due to rabies every 15 minute, and 300 people are exposed to the risk, during the same period. More than 99% rabies deaths occur in the developing countries of Asia and Africa. The disease manifests itself as a progressive fatal encephalomyelitis, and results from infection with viruses of the genus *Lyssavirus* in the family *Rhabdoviridae*. Infection is usually acquired from transcutaneous or mucosal exposure to virus-laden saliva of a rabid animal. All warm blooded animals are susceptible, though companion animals, especially dogs, constitute the major vector in most developing countries.
- Limited access to healthcare facilities and the high costs and complex schedules of rabies biologics often hamper human rabies prophylaxis in the developing countries. The cell culture vaccines, though 100% effective when combined with appropriate wound care and use of immunoglobulins, are expensive, require multiple doses over at least 3 weeks, and demand cold chain maintenance. Canine rabies control is critical in prevention of human rabies in the endemic countries; but faces greater challenges—logistics of mass vaccination drives ever-expanding population of unowned dogs, difficulties in locating the dogs for repeat doses, disrespect of vaccine cold chain, poor immune responses in malnourished and sick animals etc. A cheaper, easily producible vaccine that requires single or a few doses, and reasonably stable at room temperature would be highly desirable in endemic

countries, at least for veterinary vaccination. In this context, a DNA vaccine could be a suitable option.

2.8.5 DNA vaccines

DNA vaccines are bacterial plasmids constructed to express an encoded protein following in vivo administration and subsequent transfection of cells. Since, DNA vaccines have shown promising results in a number of trials for prophylaxis of bacterial, viral, parasitic, autoimmune and neoplastic diseases. DNA vaccines have made significant strides in veterinary practice, wherein four DNA vaccines have been licensed recently, including those against West Nile virus infection in horses (licensed in USA), Infectious Hematopoietic Necrosis Virus in Salmon (licensed in Canada), melanoma cancer of dogs (conditionally licensed in USA) and Growth Hormone Releasing Factor therapy for pigs(licensed in Australia).

Several facts explain the interest in a DNA-based rabies vaccine approach:

- Protection from rabies is commensurate with the presence of adequate amounts of virus neutralizing antibodies, principally targeted against the rabies virus glycoprotein. The tools of recombinant DNA technology allow facile cloning of the glycoprotein gene into suitable expression vectors which mediate efficient in vivo expression of glycoprotein.
- Nanogram amounts of a plasmid-encoded protein in transfected cells can generate high-affinity antibodies. In situ production of the protein within transfected cells facilitates native post-translational modifications and obviates the issues associated with protein purification in vitro.
- DNA vaccines provide unique approaches for expanding the spectrum of immune responses, such as by employing multiple or modified antigens and targeted delivery to specific cell types or locations.
- Remarkable thermal stability of plasmids (over 50°C) may be of advantage in tropical climates.
- Generic nature of the production and purification processes of plasmid vaccines reduces cost and facilitates easier product development, compared to conventional vaccines.

2.8.6 RNA or mRNA VACCINE

Number of doses required: 2 doses, intramuscular

What to know: Since no other existing licensed or approved vaccine uses this type of technology, the Messenger RNA (mRNA) variety could be mistaken for something completely new to healthcare. However, a number of mRNA vaccines have been studied in the past for illnesses and diseases including cytomegalovirus (CMV), influenza, rabies, and the Zika virus.

According to the Centers for Disease Control and Prevention (CDC): “Researchers have been studying and working with mRNA vaccines for decades. Interest has grown in these vaccines because they can be developed in a laboratory using readily available materials. This means the process can be standardised and scaled up, making vaccine development faster than traditional methods of making vaccines.”

So how does it reportedly work? The COVID-19 RNA vaccine consists of mRNA molecules made in a lab that code for parts of the SARS-CoV-2 virus – specifically the virus’ spike protein.

Once injected into the body, the mRNA instructs the cells to produce antigens – such as the spike protein mentioned – which are then detected by immune cells, triggering a response by the body’s lymphocytes.

The killer T-cells destroy the infected cells, while the B-cells and helper T-cells support antibody production. Whoever is exposed to the COVID-19 coronavirus in the future would have an immune system that recognises it, and in turn fight off the infection.

Benefits: According to the University of Cambridge’s PHG Foundation, advantages include good safety (since there are no live components, there’s no risk of the vaccine triggering disease), reliability, and that it’s relatively simple to manufacture.

Challenges: Disadvantages include unintended effects (such as an unintended immune reaction), ensuring effective delivery into the body (since free RNA in the body is quickly broken down), storage issues, plus the fact that this type of vaccine has never previously been licensed for humans.

2.8.7 DNA fingerprinting

- 3 DNA fingerprinting, also called DNA typing, DNA profiling, genotyping, or identity testing, in genetics, method of isolating and identifying variable elements within the base-pair sequence of DNA (deoxyribonucleic acid). The technique was developed in 1984 by British geneticist Alec Jeffreys, after he noticed that certain sequences of highly variable DNA (known as minisatellites), which do not contribute to the functions of genes, are repeated within genes. Jeffreys recognized that each individual has a unique pattern of minisatellites (the only exceptions being multiple individuals from a single zygote, such as identical twins). The procedure for creating a DNA fingerprint consists of first obtaining a sample of cells, such as skin, hair, or blood cells, which contain DNA. The DNA is extracted from the cells and purified. In Jeffreys's original approach, which was based on restriction fragment length polymorphism (RFLP) technology, the DNA was then cut at specific points along the strand with proteins known as restriction enzymes. The enzymes produced fragments of varying lengths that were sorted by placing them on a gel and then subjecting the gel to an electric current (electrophoresis): the shorter the fragment, the more quickly it moved toward the positive pole (anode). The sorted double-stranded DNA fragments were then subjected to a blotting technique in which they were split into single strands and transferred to a nylon sheet. The fragments underwent autoradiography in which they were exposed to DNA probes—pieces of synthetic DNA that were made radioactive and that bound to the minisatellites. A piece of X-ray film was then exposed to the fragments, and a dark mark was produced at any point where a radioactive probe had become attached. The resultant pattern of marks could then be analyzed.

3.1 Lipids

Lipids are organic substances that are insoluble in water, soluble in organic solvents, are related to fatty acids and are utilized by the living cell. They include fats, waxes, sterols, fat-soluble vitamins, mono-, di- or triglycerides, phospholipids, etc. Unlike carbohydrates, proteins, and nucleic acids, lipids are not polymeric molecules. Lipids play a great role in the cellular structure and are the chief source of energy.

“Lipids are organic compounds that contain hydrogen, carbon, and oxygen atoms, which form the framework for the structure and function of living cells.”

What are Lipids?

These organic compounds are nonpolar molecules, which are soluble only in nonpolar solvents and insoluble in water because water is a polar molecule. In the human body, these molecules can be synthesized in the liver and are found in oil, butter, whole milk, cheese, fried foods and also in some red meats. (Fig.2.11)

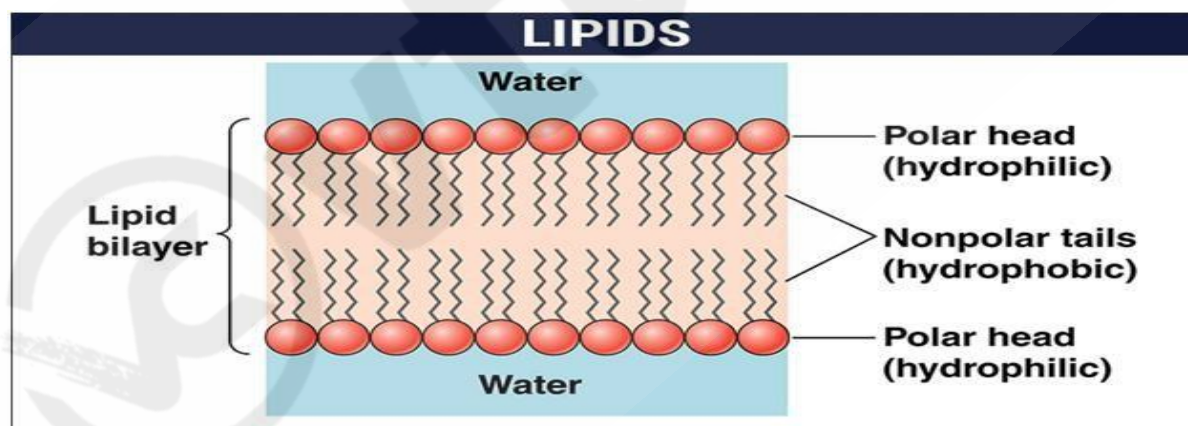


Fig. 2.11 Lipids

3.1.1 Properties of Lipids

Lipids are a family of organic compounds, composed of fats and oils. These molecules yield high energy and are responsible for different functions within the human body. Listed below are some important characteristics of Lipids.

- Lipids are oily or greasy nonpolar molecules, stored in the adipose tissue of the body.

- Lipids are a heterogeneous group of compounds, mainly composed of hydrocarbon chains.
- Lipids are energy-rich organic molecules, which provide energy for different life processes.
- Lipids are a class of compounds characterised by their solubility in nonpolar solvents and insolubility in water.
- Lipids are significant in biological systems as they form a mechanical barrier dividing a cell from the external environment known as the cell membrane.

3.1.2 Lipid Structure

Lipids are the polymers of fatty acids that contain a long, non-polar hydrocarbon chain with a small polar region containing oxygen. The lipid structure is explained in the diagram below:

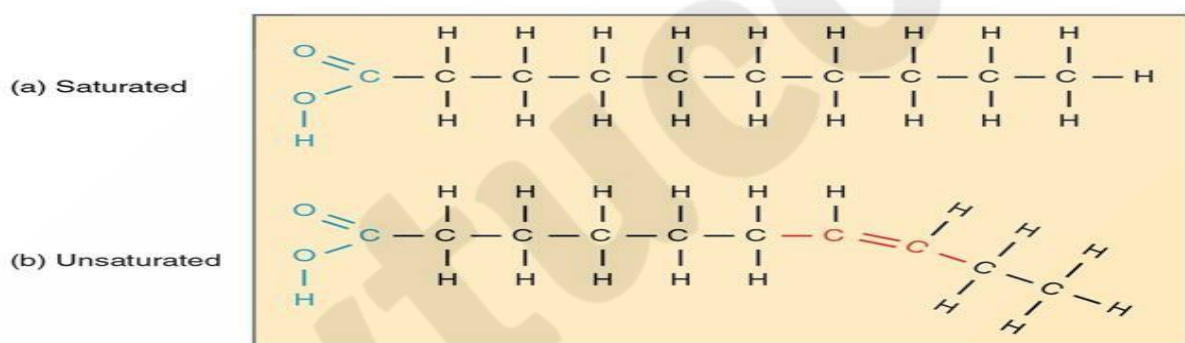


Fig. 1.12 Lipid Structure – Saturated and Unsaturated Fatty Acids

3.1.3 Classification of Lipids

Lipids can be classified into two main classes:

- Nonsaponifiable lipids
- Saponifiable lipids

A. Nonsaponifiable Lipids

A nonsaponifiable lipid cannot be disintegrated into smaller molecules through hydrolysis. Nonsaponifiable lipids include cholesterol, prostaglandins, etc

B. Saponifiable Lipids

A saponifiable lipid comprises one or more ester groups, enabling it to undergo hydrolysis in the presence of a base, acid, or **enzymes**, including waxes, triglycerides, sphingolipids and phospholipids.

Further, these categories can be divided into non-polar and polar lipids.

Nonpolar lipids, namely triglycerides, are utilized as fuel and to store energy.

Polar lipids, that could form a barrier with an external water environment, are utilized in membranes. Polar lipids comprise sphingolipids and glycerophospholipids.

Fatty acids are pivotal components of all these lipids.

1.16.4 Types of Lipids

Within these two major classes of lipids, there are numerous specific types of lipids, which are important to life, including fatty acids, triglycerides, glycerophospholipids, sphingolipids and steroids. These are broadly classified as simple lipids and complex lipids.

A.Simple Lipids

Esters of fatty acids with various alcohols.

- **Fats:** Esters of fatty acids with glycerol. Oils are fats in the liquid state
- **Waxes:** Esters of fatty acids with higher molecular weight monohydric alcohols

B.Complex Lipids

Esters of fatty acids containing groups in addition to alcohol and fatty acid.

- **Phospholipids:** These are lipids containing, in addition to fatty acids and alcohol, phosphate group. They frequently have nitrogen-containing bases and other substituents, eg, in glycerophospholipids the alcohol is glycerol and in sphingophospholipids the alcohol is sphingosine.
- **Glycolipids (glycosphingolipids):** Lipids containing a fatty acid, sphingosine and carbohydrate.
- **Other complex lipids:** Lipids such as sulfolipids and amino lipids. Lipoproteins may also be placed in this category.

C. Precursor and Derived Lipids

These include fatty acids, glycerol, steroids, other alcohols, fatty aldehydes, and ketone bodies, hydrocarbons, lipid-soluble vitamins, and hormones. Because they are uncharged, acylglycerols (glycerides), cholesterol, and cholesteryl esters are termed neutral lipids.

These compounds are produced by the hydrolysis of simple and complex lipids.

3.2 Biodiesel

- Biodiesel is a liquid biofuel obtained by chemical processes from vegetable oils or animal fats and an alcohol that can be used in diesel engines, alone or blended with diesel oil.
- ASTM (American Society for Testing and Materials) defines biodiesel as a mixture of long-chain monoalkylic esters from fatty acids obtained from renewable resources, to be used in diesel engines.
- Blends with diesel fuel are indicated as “Bx”, where “x” is the percentage of biodiesel in the blend.
- For instance, “B5” indicates a blend with 5% biodiesel and 95% diesel fuel; in consequence, B100 indicates pure biodiesel.
- Biodiesel is an alternative energy source and could be a substitute for petroleum-based diesel fuel.
- The biggest challenge modern industrial society is facing today is the decline and exhaustion of the fossil energy resources.
- The primary sources of energy that power our civilization are those fossil fuels.
- The International Energy Agency has forecasted that the world's total energy requirements will rise by half in the next 25 years.
- On the other hand, global oil product demand has been revised to 84.3 million barrels a day (mb/d) in 2006 and 85.8 mb/d in 2007
- Therefore continued use of petroleum-sourced fuels is now widely recognized as unsustainable because of depleting supplies and increasing demand.
- Oil prices have been rising steadily over the past 3 years and surged to a record high above \$60 a barrel in June 2005, sustaining a rally built on strong demand for gasoline and diesel and on concerns about refiners' ability to keep up.

- In addition, the combustion of the fossil fuels used is considered as the major factor responsible for global warming due to large-scale carbon dioxide emissions.
- One of the most prominent alternative energy resources, attracting more and more interest in recent years with the price for crude oil reaching record heights, is biodiesel, which is a possible substitute for petroleum-based diesel fuel.
- Many studies have shown that the properties of biodiesel are very close to diesel fuel.
- Therefore, biodiesel fuel can be used in diesel engines with little or no modification.
- Biodiesel has a higher cetane number than diesel fuel, no aromatics, no sulfur, and contains 10-11% oxygen by weight.
- These characteristics of biodiesel are responsible for a reduction in the emissions of carbon monoxide (CO), hydrocarbon (HC) and particulate matter (PM) in the exhaust gas compared to diesel fuel.
- However, to be a viable alternative, a biodiesel should provide a net energy gain, be economically competitive, and be producible in large quantities without reducing food s

2.10.1 Advantages of the Use of Biodiesel Some of the advantages of using biodiesel as a replacement for diesel fuel are:

- Renewable fuel, obtained from vegetable oils or animal fats.
- Low toxicity, in comparison with diesel fuel.
- Degrades more rapidly than diesel fuel, minimizing the environmental consequences of biofuel spills.
- Lower emissions of contaminants: carbon monoxide, particulate matter, polycyclic aromatic hydrocarbons, aldehydes.
- Lower health risk, due to reduced emissions of carcinogenic substances.
- No sulfur dioxide (SO₂) emissions.
- Higher flash point (100C minimum).
- May be blended with diesel fuel at any proportion; both fuels may be mixed during the fuel supply to vehicles.
- Excellent properties as a lubricant.
- It is the only alternative fuel that can be used in a conventional diesel engine, without modifications.

- Used cooking oils and fat residues from meat processing may be used as raw materials.

2.10.2 Disadvantages of the Use of Biodiesel

There are certain disadvantages of using biodiesel as a replacement for diesel fuel that must be taken into consideration:

- Slightly higher fuel consumption due to the lower calorific value of biodiesel.
- Slightly higher nitrous oxide (NO_x) emissions than diesel fuel.
- Higher freezing point than diesel fuel. This may be inconvenient in cold climates.
- It is less stable than diesel fuel, and therefore long-term storage (more than six months) of biodiesel is not recommended.
- May degrade plastic and natural rubber gaskets and hoses when used in pure form, in which case replacement with Teflon components is recommended.
- It dissolves the deposits of sediments and other contaminants from diesel fuel in storage tanks and fuel lines, which then are flushed away by the biofuel into the engine, where they can cause problems in the valves and injection systems.
- In consequence, the cleaning of tanks prior to filling with biodiesel is recommended.
- It must be noted that these disadvantages are significantly reduced when biodiesel is used in blends with diesel fuel.

2.11 Soaps and Detergents

Carboxylic acids and salts having alkyl chains longer than eight carbons exhibit unusual behavior in water due to the presence of both hydrophilic (CO₂) and hydrophobic (alkyl) regions in the same molecule. Such molecules are termed **amphiphilic** (Gk. amphi = both) or **amphipathic**. Fatty acids made up of ten or more carbon atoms are nearly insoluble in water, and because of their lower density, float on the surface when mixed with water. Unlike paraffin or other alkanes, which tend to puddle on the water's surface, these fatty acids spread evenly over an extended water surface, eventually forming a monomolecular layer in which the polar carboxyl groups are hydrogen bonded at the water interface, and the hydrocarbon chains are aligned together away from the water. This behavior is illustrated in the diagram on the right. Substances that accumulate at water surfaces and change the surface properties are called **surfactants**.

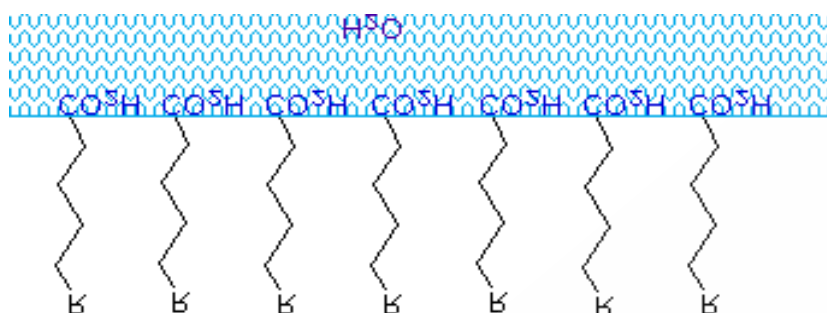
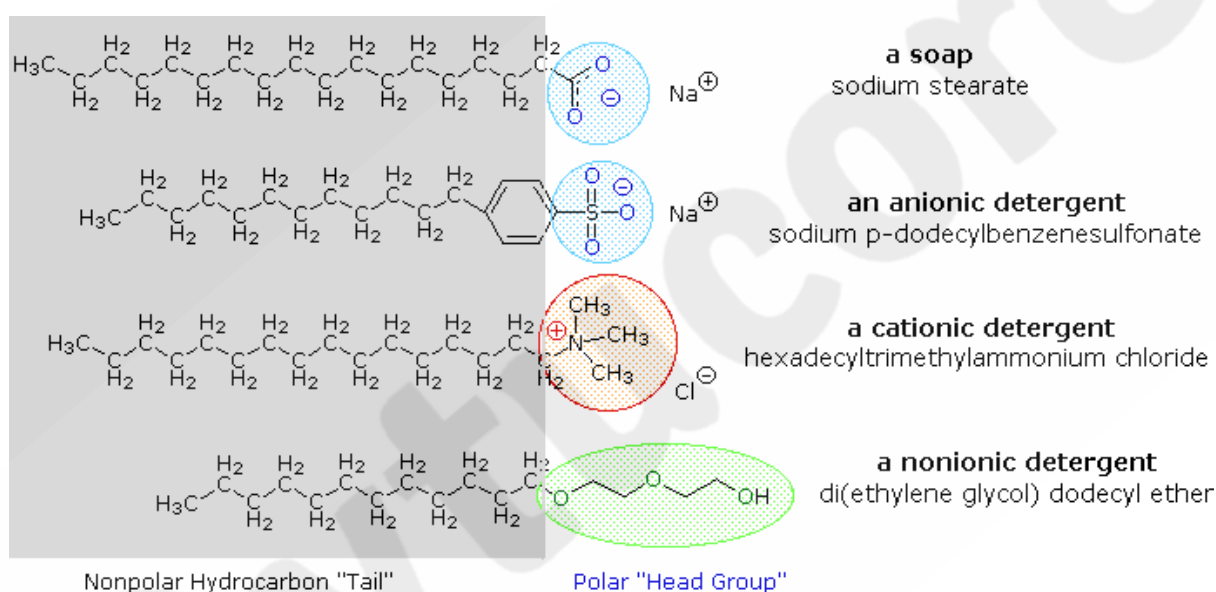


Fig. 2.13 Hydrocarbon chains



Alkali metal salts of fatty acids are more soluble in water than the acids themselves, and the amphiphilic character of these substances also make them strong surfactants. The most common examples of such compounds are soaps and detergents, four of which are shown below. Note that each of these molecules has a nonpolar hydrocarbon chain, the "tail", and a polar (often ionic) "head group". The use of such compounds as cleaning agents is facilitated by their surfactant character, which lowers the surface tension of water, allowing it to penetrate and wet a variety of materials.

Fig.2.14 Head group

Very small amounts of these surfactants dissolve in water to give a random dispersion of solute molecules. However, when the concentration is increased an interesting change occurs. The surfactant molecules reversibly assemble into polymolecular aggregates called **micelles**. By

gathering the hydrophobic chains together in the center of the micelle, disruption of the hydrogen bonded structure of liquid water is minimized, and the polar head groups extend into the surrounding water where they participate in hydrogen bonding. These micelles are often spherical in shape, but may also assume cylindrical and branched forms, as illustrated on the right. Here the polar head group is designated by a blue circle, and the nonpolar tail is a zig-zag black line.

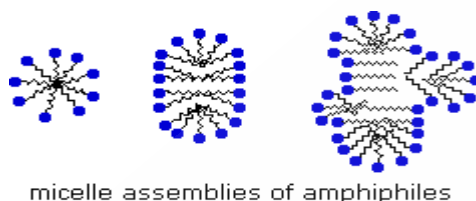


Fig.2.15 Micelle formation

An animated display of micelle formation is presented below. Notice the brownish material in the center of the three-dimensional drawing on the left. This illustrates a second important factor contributing to the use of these amphiphiles as cleaning agents. Micelles are able to encapsulate nonpolar substances such as grease within their hydrophobic center, and thus solubilize it so it is removed with the wash water. Since the micelles of anionic amphiphiles have a negatively charged surface, they repel one another and the nonpolar dirt is effectively emulsified. To summarize, **the presence of a soap or a detergent in water facilitates the wetting of all parts of the object to be cleaned, and removes water-insoluble dirt by incorporation in micelles.** If the animation has stopped, it may be restarted by clicking on it.

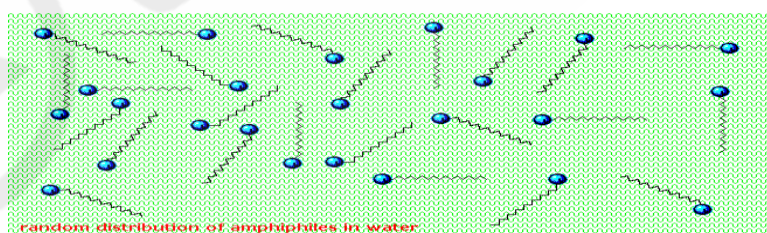


Fig.2.16 Amphiphiles in water

The oldest amphiphilic cleaning agent known to humans is soap. Soap is manufactured by the base-catalyzed hydrolysis (saponification) of animal fat (see below). Before sodium hydroxide was commercially available, a boiling solution of potassium carbonate leached from wood ashes was used. Soft potassium soaps were then converted to the harder sodium soaps by washing with salt solution. The importance of soap to human civilization is documented by history, but some problems

associated with its use have been recognized. One of these is caused by the weak acidity (pK_a ca. 4.9) of the fatty acids. Solutions of alkali metal soaps are slightly alkaline (pH 8 to 9) due to hydrolysis. If the pH of a soap solution is lowered by acidic contaminants, insoluble fatty acids precipitate and form a scum. A second problem is caused by the presence of calcium and magnesium salts in the water supply (hard water).

These divalent cations cause aggregation of the micelles, which then deposit as a dirty scum. These problems have been alleviated by the development of synthetic amphiphiles called detergents (or syndets). By using a much stronger acid for the polar head group, water solutions of the amphiphile are less sensitive to pH changes. Also the sulfonate functions used for virtually all anionic detergents confer greater solubility on micelles incorporating the alkaline earth cations found in hard water. Variations on the amphiphile theme have led to the development of other classes, such as the cationic and nonionic detergents shown above. Cationic detergents often exhibit germicidal properties, and their ability to change surface pH has made them useful as fabric softeners and hair conditioners. These versatile chemical "tools" have dramatically transformed the household and personal care cleaning product markets over the past fifty years.

2.12 Enzymes

2.12.1 Enzymes Structure

Enzymes are a linear chain of amino acids, which give rise to a three-dimensional structure. The sequence of amino acids specifies the structure, which in turn identifies the catalytic activity of the enzyme. Upon heating, the enzyme's structure denatures, resulting in a loss of enzyme activity, which typically is associated with temperature.

Compared to its substrates, enzymes are typically large with varying sizes, ranging from 62 amino acid residues to an average of 2500 residues found in fatty acid synthase. Only a small section of the structure is involved in catalysis and is situated next to the binding sites. The catalytic site and binding site together constitute the enzyme's active site. A small number of ribozymes exist which serve as an RNA-based biological catalyst. It reacts in complex with proteins.

2.12.2 Enzymes Classification

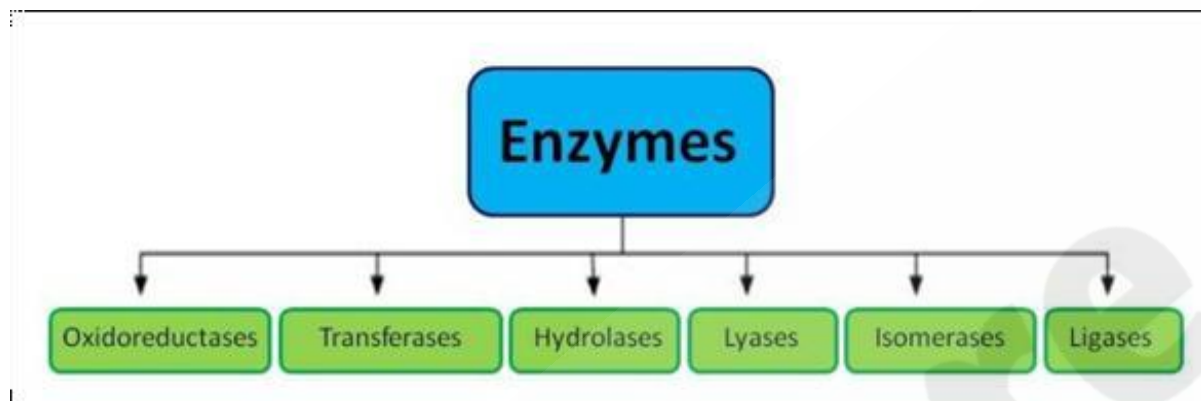


Fig.2.17 Enzymes Classification

Earlier, enzymes were assigned names based on the one who discovered them. With further research, classification became more comprehensive.

According to the International Union of Biochemists (IUB), enzymes are divided into six functional classes and are classified based on the type of reaction in which they are used to catalyze. The six kinds of enzymes are hydrolases, oxidoreductases, lyases, transferases, ligases and isomerases.

Table 2.4 Classification of enzymes

Types	Biochemical Property
Oxidoreductases	The enzyme Oxidoreductase catalyzes the oxidation reaction where the electrons tend to travel from one form of a molecule to the other.
Transferases	The Transferases enzymes help in the transportation of the functional group among acceptors and donor molecules.
Hydrolases	Hydrolases are hydrolytic enzymes, which catalyze the hydrolysis reaction by adding water to cleave the bond and hydrolyze it.
Lyases	Adds water, carbon dioxide or ammonia across double bonds or eliminate these to create double bonds.
Isomerases	The Isomerases enzymes catalyze the structural shifts present in a molecule, thus causing the change in the shape of the molecule.

Ligases	The Ligases enzymes are known to charge the catalysis of a ligation process.
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2.12.3 A biosensor

A biosensor can be defined as a “compact analytical device or unit incorporating a biological or biologically derived sensitive recognition element integrated or associated with a physio-chemical transducer”. There are three main parts of a biosensor:

- (i) the biological recognition elements that differentiate the target molecules in the presence of various chemicals,
- (ii) a transducer that converts the biorecognition event into a measurable signal, and
- (iii) a signal processing system that converts the signal into a readable form.

The molecular recognition elements include receptors, enzymes, antibodies, nucleic acids, microorganisms and lectins. The five principal transducer classes are electrochemical, optical, thermometric, piezoelectric, and magnetic. The majority of the current glucose biosensors are of the electrochemical type, because of their better sensitivity, reproducibility, and easy maintenance as well as their low cost. Electrochemical sensors may be subdivided into potentiometric, amperometric, or conductometric types. Enzymatic amperometric glucose biosensors are the most common devices commercially available and have been widely studied over the last few decades. Amperometric sensors monitor currents generated when electrons are exchanged either directly or indirectly between a biological system and an electrode.

Generally, glucose measurements are based on interactions with one of three enzymes: hexokinase, glucose oxidase (GOx) or glucose-1-dehydrogenase (GDH). The hexokinase assay is the reference method for measuring glucose using spectrophotometry in many clinical laboratories. Glucose biosensors for SMBG are usually based on the two enzyme families, GOx and GDH. These enzymes differ in redox potentials, cofactors, turnover rate and selectivity for glucose. GOx is the standard enzyme for biosensors; it has a relatively higher selectivity for glucose. GOx is easy to obtain, cheap, and can withstand greater extremes of pH, ionic strength, and temperature than many other enzymes, thus allowing less stringent conditions during the manufacturing process and relatively relaxed storage norms for use by lay biosensor users.

The basic concept of the glucose biosensor is based on the fact that the immobilized GOx catalyzes the oxidation of β -D-glucose by molecular oxygen producing gluconic acid and hydrogen peroxide. In order to work as a catalyst, GOx requires a redox cofactor—flavin adenine dinucleotide (FAD). FAD works as the initial electron acceptor and is reduced to FADH₂.



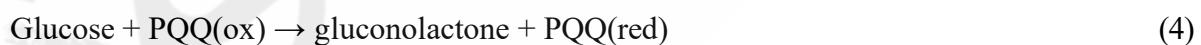
The cofactor is regenerated by reacting with oxygen, leading to the formation of hydrogen peroxides.



Hydrogen peroxide is oxidized at a catalytic, classically platinum (Pt) anode. The electrode easily recognizes the number of electron transfers, and this electron flow is proportional to the number of glucose molecules present in blood.



Three general strategies are used for the electrochemical sensing of glucose; by measuring oxygen consumption, by measuring the amount of hydrogen peroxide produced by the enzyme reaction or by using a diffusible or immobilized mediator to transfer the electrons from the GOx to the electrode. The number and types of GDH-based amperometric biosensors have been increasing recently. The GDH family includes GDH-pyrroquinolinequinone (PQQ) and GDH-nicotinamide-adenine dinucleotide (NAD). The enzymatic reaction of GDH is independent of the dissolved oxygen. The quinoprotein GDH recognition element uses PQQ as a cofactor.



This mechanism requires neither oxygen nor NAD⁺. GDH-PQQ is a particularly efficient enzyme system, with a rapid electron transfer rate, but it is relatively expensive.

GDH with NAD as a cofactor produces NADH rather than H₂O₂. NAD is a major electron acceptor in the oxidation of glucose, during which the nicotinamide ring of NAD⁺ accepts a hydrogen ion and two electrons, equivalent to a hydride ion. The reduced form of this carrier generated in this reaction is called NADH, which can be electrochemically oxidized.

